

Fig. 2: A single-side PCB layout of the temperature-based fan-speed controller, shown in actual size

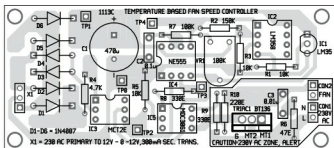


Fig. 3: Component layout for the PCB

TRIAC1 is used to control the speed of the fan by adjusting the RMS value of the AC voltage across the fan.

connected with a snubber circuit built around resistor R6 and capacitor C3, to protect it from high-voltage transients.

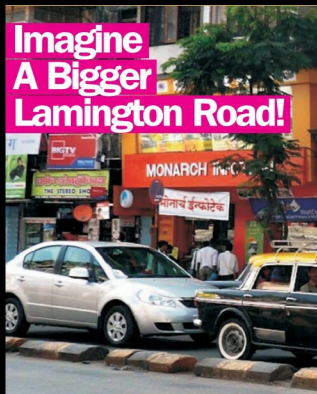
The firing pulses for TRIAC1 are produced by a monostable multivibrator. The width of these firing pulses varies in accordance with the output voltage of temperature sensor IC1. At a higher temperature, the width of the firing pulses will be more and, hence, the RMS value of the AC voltage will increase. As a result, the fan will rotate at a higher speed. TRIAC1 is

Construction and testing

A single-side PCB for the fan-speed controller is shown in actual size in Fig. 2 and its component layout is shown in Fig. 3. After assembling the circuit on the recommended PCB and enclosing it in a suitable case, you can calibrate the system as per your requirements, using VR1. Change VR1 so that, at room temperature, you get the normal speed. As the temperature increases, the speed increases beyond the speed set at room temperature.

To test if the circuit is functioning properly, check the input power supply at TP1 with respect to TP0. Trigger pulses for IC4 can be observed at TP2 using an oscilloscope. Also verify the pulse-width-modulated output of IC4 at TP3. The amplified output of temperature sensor IC1 can be checked at TP4. ●

The author is a final year B.Tech (Electrical and Electronics Engineering) student at MG University. His areas of interest include embedded systems and power electronics



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